

FSOC VIRTUAL CAMERA TRACKING SYSTEM

TARGET: TGT-01

Coarse Alignment - Point-in-Time Performance Report

T = 2.00 s

1. TEST CONFIGURATION

Scenario

Straight Line

Target

TGT-01

Camera

640 x 480

FOV

4.0 deg x 3.0 deg

Update Rate

30.0 Hz

Noise: None 10.00 | Camera jitter: OFF 5.00 px/frame | Platform motion: OFF 10.00 px | Turbulence: OFF 10.00 px

2. PERFORMANCE

Processing FPS

123.8

>= 20 FPS

Average Tracking Error

1.93 px

Maximum Tracking Error

4.00 px

<= 10 px

Acquisition Time

0.60 s

<= 2 s

Re-acquisition

- s

<= 1 s

Lock Retention

71.67 %

RMSE

2.13 px

Target Loss

0.00 %

Avg Processing

8.08 ms

Re-acquisitions

0

3. PS REQUIREMENT CHECK

Parameter	Measured	Requirement	Status
Update Rate	30.0 Hz	>= 20 Hz	PASS
Processing Speed	123.8 FPS	>= 20 FPS	PASS
Acquisition Time	0.60 s	<= 2 s	PASS
Tracking Error (Max)	4.00 px	<= 10 px	PASS
Target Loss	0.00 %	< 5 %	PASS
Re-acquisition	- s	<= 1 s	-

4. VISUAL SNAPSHOT (T = 2.00 s)

RADAR



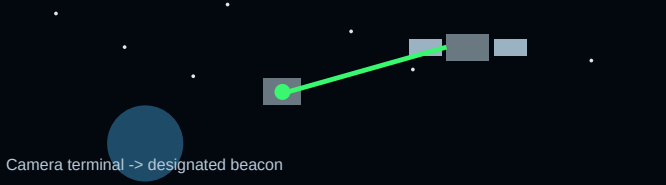
Target / Boresight / Prediction

VIRTUAL CAMERA



Target TGT-01 • Error 2.00 px

3D DIGITAL TWIN



Camera terminal -> designated beacon

State: TRACKING | Pan: -0.12 deg | Tilt: 0.00 deg | AI: 0.79

5. SYSTEM SUMMARY

HANDOFF READY - COARSE ALIGNMENT GATE SATISFIED

Designated target TGT-01 at 2.00 s. Detection, AI verification, Kalman prediction and pan/tilt control are reported from the recorded telemetry snapshot.